KDD Cup 2015 → FSLSM Mapping & K=3 Clustering

1. FSLSM – KDD Cup 2015 Feature Mapping

Dimension	Pole	Log Event(s)	Object Category
Input	Visual	video, access	video
Input	Verbal	access, wiki, discussion	html, discussion, wiki
Processing	Active	problem, discussion	problem, discussion
Processing	Reflective	video, access (high dwell)	video, html
Perception	Sensing	problem (high retry)	problem
Perception	Intuitive	wiki, navigate (broad)	html, chapter, sequential
Understanding	Sequential	navigate (low variance)	chapter, sequential, vertical
Understanding	Global	navigate (jumps), page_close	chapter, sequential, vertical

Overall Experimental Steps

1. **KDD Cup 2015 Interaction Logs** – Use the learner interaction logs containing activities such as video, access, problem, discussion, wiki, navigate, and page-close events.
2. **FSLSM-Based Event Mapping** – Map each interaction event to the appropriate FSLSM dimension and pole: Input, Processing, Perception, and Understanding.
3. **FSLSM-Oriented Feature Extraction** – Convert the mapped interactions into numerical behavioural features such as interaction frequency, dwell time, retries, and navigation patterns for each FSLSM pole.
4. **Learner-Level Feature Construction** – Aggregate the extracted interaction features for each learner to create a single learner-level feature vector.
5. **Feature Normalization** – Normalize the learner features so that features with different scales do not disproportionately influence the subsequent representation learning and clustering.
6. **Autoencoder Training** – Train an Autoencoder on the normalized learner features to learn a compact representation of each learner's interaction behaviour.
7. **Latent Representation Extraction** – Extract the encoded latent representation from the Autoencoder and use it as the input for clustering.
8. **K-Means Clustering (K=3)** – Apply K-Means with three clusters to group learners according to their latent interaction patterns.

9. **DEC Clustering (K=3)** – Apply Deep Embedded Clustering with three clusters to jointly refine the latent representation and learner clusters.
10. **Clustering Evaluation** – Compare K-Means and DEC using Silhouette Score, Davies–Bouldin Index, and Calinski–Harabasz Index to determine which method produces better-separated clusters.
11. **Selection of Best Clustering Method** – Select the clustering approach that provides the most meaningful and well-separated three-cluster solution.
12. **Three-Class Learning-Style Label Generation** – Interpret the three clusters for each FSLSM dimension as three learning-style categories representing the two poles and a balanced category, such as **Visual–Balanced–Verbal**.
13. **Full-Course Learning-Style Label** – Aggregate interactions across all course weeks and generate one overall three-class learning-style label for each of the four FSLSM dimensions.
14. **Week-1 Learning-Style Label** – Use only the learner's Week-1 interactions to generate an initial learning-style profile representing the learner's early behavioural characteristics.
15. **Last-Week Learning-Style Label** – Use the interactions from the final available week to generate a later learning-style profile and examine changes in the learner's behavioural style.
16. **Baseline Dropout Prediction** – Predict learner dropout using the original behavioural features without incorporating any learning-style labels.
17. **Dropout Prediction with Full-Course Learning Style** – Add the full-course three-class learning-style labels to the behavioural features and evaluate whether they improve dropout prediction.
18. **Dropout Prediction with Week-1 and Last-Week Learning Style** – Add both the Week-1 and Last-Week learning-style labels to examine whether temporal learning-style information improves dropout prediction.
19. **Early-Warning Dropout Prediction** – Progressively use Week 1, Weeks 1–2, Weeks 1–3, and so on to generate learning-style information and predict subsequent dropout at each stage.
20. **Performance Comparison** – Compare all dropout prediction settings using **Accuracy, Precision, Recall, F1-score, and preferably ROC-AUC**, and determine whether learning-style information improves prediction and how early it becomes useful.